EXTENSION 1

Stage 6

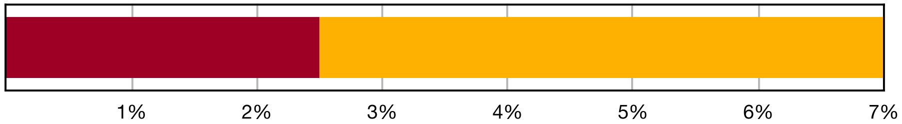
Trigonometry (Ext1), T3 Trig Equations (Y12)

Auxiliary Angles (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 57 minutes *(based on allocation of 1.5 minutes per mark)*

T3 Trig Equations



Auxiliary Angles
Identities, Equations and 't' Formulae

***SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

HISTORICAL CONTRIBUTION

- *T3 Trig Equations* is easily the biggest sub-topic within Trigonometry and has contributed an impressive average of 7.0% per new syllabus Ext1 exam since its introduction in 2020.
- This topic has been split into two sub-categories for analysis purposes which are: *1-Auxiliary Angles (2.5%) and 2-Identities, Equations and 't' Formulae (4.5%).*
- This analysis looks at *Auxiliary Angles.*

HSC ANALYSIS - What to expect and common pitfalls

- *Auxiliary Angles (2.5%)* have been tested in a dedicated longer answer question in both the 2022 and 2020 new syllabus exams (omitted in 2023, 2021). This continues a longer trend of this question type being asked in a clear majority of papers (6 times in the last decade).
- Mark allocations are more often in the substantial 3-4 range, making this a fertile area for scoring well, although we note 2014 and 2017 examined this topic via multiple choice.
- Efficiency in answering auxiliary angle questions requires students to have a clear and precise answer structure in their mind before they start. We highly recommend revision that develops this structured approach which is illustrated in the worked solutions.
- Most questions in this area test around the band 4 difficulty level, although *2010 Ext1 4b* proved more challenging and deserves close attention.

Questions

1. Trigonometry, EXT1 T3 2014 HSC 2 MC

Which expression is equal to $\cos x - \sin x$?

A. $\sqrt{2}\cos\left(x + \dfrac{\pi}{4}\right)$

B. $\sqrt{2}\cos\left(x - \dfrac{\pi}{4}\right)$

C. $2\cos\left(x + \dfrac{\pi}{4}\right)$

D. $2\cos\left(x - \dfrac{\pi}{4}\right)$

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2. Trigonometry, EXT1 T3 2017 HSC 4 MC

What is the value of $\tan\alpha$ when the expression $2\sin x - \cos x$ is written in the form $\sqrt{5}\sin(x - \alpha)$?

A. -2

B. $-\dfrac{1}{2}$

C. $\dfrac{1}{2}$

D. 2

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3. Trigonometry, EXT1 T3 2024 HSC 10 MC

For real numbers a and b, where $a \neq 0$ and $b \neq 0$, we can find numbers α, β, γ, δ and R such that $a\cos x + b\sin x$ can be written in the following 4 forms:

$R\sin(x + \alpha)$

$R\sin(x - \beta)$

$R\cos(x + \gamma)$

$R\cos(x - \delta)$

where $R > 0$ and $0 < \alpha, \beta, \gamma, \delta < 2\pi$.

What is the value of $\alpha + \beta + \gamma + \delta$?

A. 0

B. π

C. 2π

D. 4π

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4. Trigonometry, EXT1 T3 2015 HSC 11d

Express $5\cos x - 12\sin x$ in the form $A\cos(x + \alpha)$, where $0 \leq \alpha \leq \dfrac{\pi}{2}$. *(2 marks)*

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5. Trigonometry, EXT1 T3 2018 HSC 11c

Write $\sqrt{3}\sin x + \cos x$ in the form $R\sin(x+\alpha)$ where $R > 0$ and $0 \le \alpha \le \dfrac{\pi}{2}$. *(2 marks)*

6. Trigonometry, EXT1 T3 2013 HSC 12a

i. Write $\sqrt{3}\cos x - \sin x$ in the form $2\cos(x+\alpha)$, where $0 < \alpha < \dfrac{\pi}{2}$. *(1 mark)*

ii. Hence, or otherwise, solve $\sqrt{3}\cos x = 1 + \sin x$, where $0 < x < 2\pi$. *(2 marks)*

7. Trigonometry, EXT1 T3 2004 HSC 2d

i. Write $8\cos x + 6\sin x$ in the form $A\cos(x - \alpha)$, where $A > 0$ and $0 \le \alpha \le \dfrac{\pi}{2}$. *(2 marks)*

ii. Hence, or otherwise, solve the equation $8\cos x + 6\sin x = 5$ for $0 \le x \le 2\pi$.
Give your answers correct to three decimal places. *(2 marks)*

8. Trigonometry, EXT1 T3 2009 HSC 2b

i. Express $3\sin x + 4\cos x$ in the form $A\sin(x+\alpha)$ where $0 \le \alpha \le \dfrac{\pi}{2}$. *(2 marks)*

ii. Hence, or otherwise, solve $3\sin x + 4\cos x = 5$ for $0 \le x \le 2\pi$.
Give your answer, or answers, correct to two decimal places. *(2 marks)*

9. Trigonometry, EXT1 T3 2020 HSC 11d

By expressing $\sqrt{3}\sin x + 3\cos x$ in the form $A\sin(x+a)$, solve $\sqrt{3}\sin x + 3\cos x = \sqrt{3}$, for $0 \le x \le 2\pi$. *(4 marks)*

10. Trigonometry, EXT1 T3 2022 HSC 11e

Express $\sqrt{3}\sin(x) - 3\cos(x)$ in the form $R\sin(x+\alpha)$. *(3 marks)*

11. Trigonometry, EXT1 T3 EQ-Bank 4

The current flowing through an electrical circuit can be modelled by the function

$$f(t) = 6\sin 0.05t + 8\cos 0.05t, \quad t \ge 0$$

i. Express the function in the form $f(t) = A\sin(at+b), \quad \text{for} \quad 0 \le b \le \dfrac{\pi}{2}$. *(2 marks)*

ii. Find the time at which the current first obtains it maximum value. *(1 mark)*

iii. Sketch the graph of $f(t)$. Clearly show its range and label the coordinates of its first maximum value.
Do not label x-intercepts. *(1 mark)*

12. Trigonometry, EXT1 T3 2010 HSC 4b

i. Express $2\cos\theta + 2\cos\left(\theta + \dfrac{\pi}{3}\right)$ in the form $R\cos(\theta+\alpha)$,

where $R > 0$ and $0 < \alpha < \dfrac{\pi}{2}$. *(3 marks)*

ii. Hence, or otherwise, solve $2\cos\theta + 2\cos\left(\theta + \dfrac{\pi}{3}\right) = 3$,

for $0 < \theta < 2\pi$. *(2 marks)*

13. Trigonometry, EXT1 T3 EQ-Bank 2

A particular energy wave can be modelled by the function

$$f(t) = \sqrt{5}\sin 0.2t + 2\cos 0.2t, \quad t \in [0, 50]$$

i. Express this function in the form $f(t) = R\sin(nt - \alpha), \quad \alpha \in [0, 2\pi]$. *(2 marks)*

ii. Find the time the wave first attains its maximum value. Give your answer to one decimal place. *(2 marks)*

Worked Solutions

1. Trigonometry, EXT1 T3 2014 HSC 2 MC

$$R\cos(x+\alpha) = R\cos x\cos\alpha - R\sin x\sin\alpha$$
$$\therefore\ R\cos x\cos\alpha - R\sin x\sin\alpha = \cos x - \sin x$$
$$R\cos\alpha = 1, \quad R\sin\alpha = 1$$
$$R^2 = 1^2 + 1^2$$
$$R = \sqrt{2}$$
$$\therefore\ \sqrt{2}\cos\alpha = 1$$
$$\cos\alpha = \frac{1}{\sqrt{2}}$$
$$\alpha = \frac{\pi}{4}$$
$$\therefore\ \sqrt{2}\cos\left(x + \frac{\pi}{4}\right) = \cos x - \sin x$$
$$\Rightarrow A$$

2. Trigonometry, EXT1 T3 2017 HSC 4 MC

Using $\sqrt{5}\sin(x-\alpha) = \sqrt{5}\sin x\cos\alpha - \sqrt{5}\cos x\sin\alpha$,

$$\Rightarrow \sqrt{5}\sin x\cos\alpha - \sqrt{5}\cos x\sin\alpha = 2\sin x - \cos x$$
$$\Rightarrow \sqrt{5}\cos\alpha = 2,\ \sqrt{5}\sin\alpha = 1$$
$$\frac{\sin\alpha}{\cos\alpha} = \frac{1}{2}$$
$$\therefore \tan\alpha = \frac{1}{2}$$
$$\Rightarrow C$$

Worked Solutions

3. Trigonometry, EXT1 T3 2024 HSC 10 MC

$a\sin x + b\cos x$ can be written 4 ways.

Consider the case:

$\cos x + \sin x$ where $a = b = 1,\ \ R = \sqrt{2}$

$$\sqrt{2}\sin(x+\alpha) = \sqrt{2}(\sin x\cos\alpha + \cos x\sin\alpha)$$
$$= \sqrt{2}\left(\sin x\cdot\frac{1}{\sqrt{2}} + \cos x\cdot\frac{1}{\sqrt{2}}\right)$$

$$\cos\alpha^+, \sin\alpha^+ \ \Rightarrow\ \alpha = \frac{\pi}{4}$$

Similarly for other 3 cases:

$$\sqrt{2}\sin(x-\beta):\ \cos\beta^+, \sin\beta^- \ \Rightarrow\ \beta = \frac{7\pi}{4}$$
$$\sqrt{2}\cos(x+\gamma):\ \cos\gamma^+, \sin\gamma^- \ \Rightarrow\ \gamma = \frac{7\pi}{4}$$
$$\sqrt{2}\cos(x-\delta):\ \cos\delta^+, \sin\delta^+ \ \Rightarrow\ \delta = \frac{\pi}{4}$$
$$\therefore \alpha + \beta + \gamma + \delta = 4\pi$$
$$\Rightarrow D$$

4. Trigonometry, EXT1 T3 2015 HSC 11d

$$5\cos x - 12\sin x = A\cos(x + \alpha)$$
$$= A\cos x\cos\alpha - A\sin x\sin\alpha$$

$$\therefore A\cos\alpha = 5, \quad A\sin\alpha = 12$$
$$A^2 = 5^2 + 12^2 = 169$$
$$A = 13$$
$$\Rightarrow 13\cos\alpha = 5$$
$$\cos\alpha = \frac{5}{13}$$
$$\alpha = \cos^{-1}\left(\frac{5}{13}\right) \approx 1.176\ldots \text{ radians}$$

$$\therefore 5\cos x - 12\sin x = 13\cos(x + 1.176\ldots)$$

5. Trigonometry, EXT1 T3 2018 HSC 11c

$$R\sin(x+\alpha) = \sqrt{3}\sin x + \cos x$$
$$R\sin x\cos\alpha + R\cos x\sin\alpha = \sqrt{3}\sin x + \cos x$$

Equating coefficients:

$\Rightarrow R\cos\alpha = \sqrt{3}$

$\Rightarrow R\sin\alpha = 1$

$$R^2 = \left(\sqrt{3}\right)^2 + 1^2 = 4$$

$\therefore R = 2$

$$\Rightarrow 2\sin\alpha = 1$$
$$\sin\alpha = \frac{1}{2}$$
$$\alpha = \frac{\pi}{6} \quad \left(0 \le \alpha \le \frac{\pi}{2}\right)$$

$$\therefore \sqrt{3}\sin x + \cos x = 2\sin\left(x + \frac{\pi}{6}\right).$$

6. Trigonometry, EXT1 T3 2013 HSC 12a

i. Write $\sqrt{3}\cos x - \sin x$ in form

$$2\cos(x+\alpha), \quad 0 < \alpha < \frac{\pi}{2}$$

$$2(\cos x\cos\alpha - \sin x\sin\alpha) = \sqrt{3}\cos x - \sin x$$
$$\cos x\cos\alpha - \sin x\sin\alpha = \frac{\sqrt{3}}{2}\cos x - \frac{1}{2}\sin x$$

$$\Rightarrow \cos\alpha = \frac{\sqrt{3}}{2}$$
$$\Rightarrow \sin\alpha = \frac{1}{2}$$
$$\alpha = \frac{\pi}{6}$$

$$\therefore\ 2\cos\left(x + \frac{\pi}{6}\right) = \sqrt{3}\cos x - \sin x$$

ii. Solve $\sqrt{3}\cos x = 1 + \sin x, \quad 0 < x < 2\pi$

$$\sqrt{3}\cos x - \sin x = 1$$
$$2\cos\left(x + \frac{\pi}{6}\right) = 1 \quad \text{(from part (i))}$$
$$\cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$$
$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Since cos is positive in $1^{\text{st}}/4^{\text{th}}$ quadrants

$$x + \frac{\pi}{6} = \frac{\pi}{3},\ 2\pi - \frac{\pi}{3} \qquad (0 < x < 2\pi)$$
$$\therefore\ x = \frac{\pi}{6},\ \frac{3\pi}{2}$$

7. Trigonometry, EXT1 T3 2004 HSC 2d

i. Write $8\cos x + 6\sin x$ in the form $A\cos(x - \alpha)$

$$A(\cos x \cos \alpha + \sin x \sin \alpha) = 8 \cos x + 6 \sin x$$

$$(\cos x \cos \alpha + \sin x \sin \alpha) = \frac{8}{A} \cos x + \frac{6}{A} \sin x$$

$$\Rightarrow \cos \alpha = \frac{8}{A}$$

$$\Rightarrow \sin \alpha = \frac{6}{A}$$

$$\left(\frac{8}{A}\right)^2 + \left(\frac{6}{A}\right)^2 = 1$$

$$8^2 + 6^2 = A^2$$

$$\therefore A = \sqrt{100}$$

$$= 10$$

$$\Rightarrow \cos \alpha = \frac{8}{10}$$

$$\therefore \alpha = 0.6435\ldots \text{ radians}$$

$$\therefore 8\cos x + 6\sin x = 10\cos(x - 0.6435\ldots)$$

ii.

$$8 \cos x + 6 \sin x = 5$$

$$10 \cos\ (x - 0.6435\ldots) = 5$$

$$\cos\ (x - 0.6435\ldots) = \frac{1}{2}$$

$$x - 0.6435\ldots = \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3}$$

$$x = \frac{\pi}{3} + 0.6435\ldots \text{ or } \frac{5\pi}{3} + 0.6435\ldots$$

$$= 1.691 \text{ or } 5.879 \text{ radians (to 3 d.p.)}$$

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8. Trigonometry, EXT1 T3 2009 HSC 2b

i. $A\sin(x + \alpha) = 3\sin x + 4\cos x$

$$A\sin x\cos\alpha + A\cos x\sin\alpha = 3\sin x + 4\cos x$$

$$\Rightarrow A\cos\alpha = 3 \qquad A\sin\alpha = 4$$

$$A^2 = 3^2 + 4^2 = 25$$

$$\therefore A = 5$$

$$\Rightarrow 5\cos\alpha = 3$$

$$\cos\alpha = \frac{3}{5}$$

$$\alpha = \cos^{-1}\left(\frac{3}{5}\right) = 0.9272\ldots \text{ radians}$$

$$\therefore\ 3\sin x + 4\cos x = 5\sin\left(x + \cos^{-1}\left(\frac{3}{5}\right)\right)$$

ii. $3\sin x + 4\cos x = 5$

$$5\sin(x + \alpha) = 5$$

$$\sin(x + \alpha) = 1$$

$$x + \alpha = \frac{\pi}{2}, \frac{5\pi}{2}, \ldots$$

$$x = \frac{\pi}{2} - 0.9272\ldots \qquad (0 \leq x \leq 2\pi)$$

$$= 0.6435\ldots$$

$$= 0.64 \text{ radians (2 d.p.)}$$

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9. Trigonometry, EXT1 T3 2020 HSC 11d

$A\sin(x+\alpha) = \sqrt{3}\sin x + 3\cos x$

$A\sin x\cos\alpha + A\cos x\sin\alpha = \sqrt{3}\sin x + 3\cos x$

Equating co-efficients:

$\Rightarrow A\cos\alpha = \sqrt{3}$

$\Rightarrow A\sin\alpha = 3$

$A^2 = \left(\sqrt{3}\right)^2 + 3^2 = 12$

$\therefore A = \sqrt{12}$

$\dfrac{A\sin\alpha}{A\cos\alpha} = \dfrac{3}{\sqrt{3}}$

$\tan\alpha = \sqrt{3}$

$\alpha = \dfrac{\pi}{3}$

$\sqrt{3}\sin x + 3\cos x = \sqrt{3}$

$\sqrt{12}\sin\left(x + \dfrac{\pi}{3}\right) = \sqrt{3}$

$\sin\left(x + \dfrac{\pi}{3}\right) = \dfrac{1}{2}$

$x + \dfrac{\pi}{3} = \dfrac{5\pi}{6}, \dfrac{13\pi}{6}$

$\therefore x = \dfrac{\pi}{2}, \dfrac{11\pi}{6} \quad (0 \le x \le 2\pi)$

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10. Trigonometry, EXT1 T3 2022 HSC 11e

$R\sin(x+\alpha) = \sqrt{3}\sin(x) - 3\cos(x)$

$R\sin(x)\cos\alpha + R\cos(x)\sin\alpha = \sqrt{3}\sin(x) - 3\cos(x)$

Equating coefficients:

$R\cos\alpha = \sqrt{3}, \quad R\sin\alpha = -3$

$R^2 = \left(\sqrt{3}\right)^2 + (-3)^2 = 12$

$\therefore R = \sqrt{12} = 2\sqrt{3}$

Given $\sin\alpha < 0$ and $\cos\alpha > 0$

$\Rightarrow \alpha$ is in the 4th quadrant

$\sqrt{12}\cos\alpha = \sqrt{3}$

$\cos\alpha = \dfrac{\sqrt{3}}{\sqrt{12}} = \dfrac{1}{2}$

$\therefore \alpha = -\dfrac{\pi}{3} \quad (-\pi \le \alpha \le \pi)$

$\therefore \sqrt{3}\sin(x) - 3\cos(x) = 2\sqrt{3}\sin\left(x - \dfrac{\pi}{3}\right)$

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11. Trigonometry, EXT1 T3 EQ-Bank 4

i. $f(t) = 6\sin 0.05t + 8\cos 0.05t$

$$\begin{aligned} A\sin(at+b) &= A\sin(0.05t+b) \\ &= A\sin(0.05t)\ \cos(b) + A\cos(0.05t)\ \sin(b) \end{aligned}$$

$$\Rightarrow A\cos(b) = 6,\ \ A\sin(b) = 8$$

$$\begin{aligned} A^2 &= 6^2 + 8^2 \\ A &= 10 \end{aligned}$$

$$\Rightarrow 10\cos(b) = 6$$

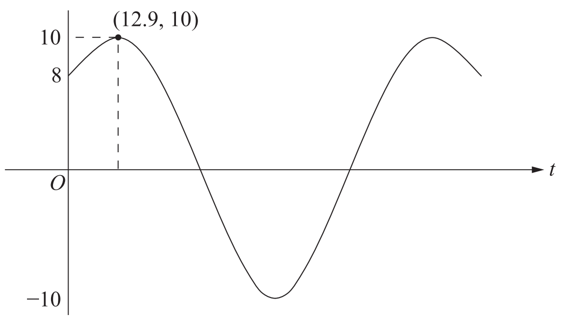
$$\begin{aligned} \cos(b) &= \frac{6}{10} \\ b &= \cos^{-1} 0.06 \approx 0.927 \text{ radians} \end{aligned}$$

$$\therefore f(t) = 10\sin(0.05t + 0.927)$$

ii. Max occurs at $\sin(0.05t + 0.927) = \sin\dfrac{\pi}{2}$

$$\begin{aligned} 0.05t + 0.927 &= \frac{\pi}{2} \\ 0.05t &= 0.643\ldots \\ \therefore t &= 12.87 \\ &= 12.9 \quad (\text{to 1 d.p.}) \end{aligned}$$

iii.



12. Trigonometry, EXT1 T3 2010 HSC 4b

i. $2\cos\theta + 2\cos\left(\theta + \frac{\pi}{3}\right)$

$$= 2\cos\theta + 2\left(\cos\theta\cos\left(\frac{\pi}{3}\right) - \sin\theta\sin\left(\frac{\pi}{3}\right)\right)$$

$$= 2\cos\theta + 2\cos\theta \times \frac{1}{2} - 2\sin\theta \times \frac{\sqrt{3}}{2}$$

$$= 2\cos\theta + \cos\theta - \sqrt{3}\sin\theta$$

$$= 3\cos\theta - \sqrt{3}\sin\theta$$

$$R\cos(\theta + \alpha) = R\cos\theta\cos\alpha - R\sin\theta\sin\alpha$$

$$R\cos\alpha = 3 \qquad R\sin\alpha = \sqrt{3}$$

$$\cos\alpha = \frac{3}{R} \qquad \sin\alpha = \frac{\sqrt{3}}{R}$$

$$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

$$\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6} \qquad \left(0 < \alpha < \frac{\pi}{2}\right)$$

$$R^2 = 3^2 + \left(\sqrt{3}\right)^2$$

$$= 9 + 3$$

$$R = \sqrt{12} = 2\sqrt{3}$$

$$\therefore\ 2\cos\theta + 2\cos\left(\theta + \frac{\pi}{3}\right) = 2\sqrt{3}\cos\left(\theta + \frac{\pi}{6}\right)$$

ii. $2\cos\theta + 2\cos\left(\theta + \frac{\pi}{3}\right) = 3$

$$2\sqrt{3}\cos\left(\theta + \frac{\pi}{6}\right) = 3$$

$$\cos\left(\theta + \frac{\pi}{6}\right) = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

Since cos is positive in 1st and 4th quadrants,

$$\theta + \frac{\pi}{6} = \frac{\pi}{6},\ 2\pi - \frac{\pi}{6}$$

$$\therefore \theta = \frac{5\pi}{3} \qquad (0 < \theta < 2\pi)$$

♦ *Mean mark part (ii) 49%*
MARKER'S COMMENT: Many students did not check their answers against the stated domain for θ.

13. Trigonometry, EXT1 T3 EQ-Bank 2

i. $f(t) = \sqrt{5}\sin 0.2t + 2\cos 0.2t$

$$R\sin(nt - \alpha) = R\sin(0.2t - \alpha)$$

$$= R\sin 0.2t\cos\alpha - R\cos 0.2t\sin\alpha$$

$$\Rightarrow R\cos\alpha = \sqrt{5}, \quad R\sin\alpha = -2$$

$$R^2 = \left(\sqrt{5}\right)^2 + (-2)^2 = 9$$

$$R = 3$$

$$\cos\alpha = \frac{\sqrt{5}}{3},\ \sin\alpha = -\frac{2}{3}$$

$\Rightarrow \alpha$ is in 4th quadrant

$$\text{Base angle} = \cos^{-1}\left(\frac{\sqrt{5}}{3}\right) = 0.7297$$

$$\therefore \alpha = 2\pi - 0.7297$$

$$= 5.553\ldots$$

$$\therefore f(t) = 3\sin(0.2t - 5.553)$$

ii. Max value occurs when $\sin(0.2t - 5.553) = 1$

$$0.2t - 5.553 = \frac{\pi}{2}$$

$$0.2t = 7.124\ldots$$

$$t = 35.62\ldots$$

Test if $t > 0$ for:

$$0.2t - 5.553 = -\frac{3\pi}{2}$$

$$0.2t = 0.8406$$

$$t = 4.20\ldots$$

$\therefore$ Time of 1st maximum: $t = 4.2$